

BIBEK POUDEL

Artesia, CA | (909) 529-2588 | bibekpoudel2056@gmail.com | bibekpoudel.com | LinkedIn

FULL-STACK AI PRODUCT ENGINEER

Full-stack AI product engineer with 3.5 years of full-time platform engineering plus enterprise AI internship experience at JPMorgan Chase. Creator of **EESA.AI**, a live multi-tenant AI agent platform, with peer-reviewed ML publications. Builds AI agent systems, RAG-backed workflows, secure APIs, and cloud-native products with a focus on architecture and production readiness.

TECHNICAL SKILLS

AI / LLM Systems: OpenAI API, LangChain, LangGraph, RAG, embeddings, semantic search, FAISS, Pinecone, Qdrant

Agent Platforms: MCP, OpenAPI 3.0-to-MCP generation, LLM routing, tool-capability gating, workflow automation, plugin marketplaces

Full-Stack: TypeScript, Java, Python, JavaScript, Next.js, React, Node.js, Express.js, Spring Boot, FastAPI, REST, GraphQL, WebSockets

Architecture: Multi-tenant SaaS, RBAC, tenant-scoped workflows, modular app systems, typed API clients, distributed systems

Data / Cloud: PostgreSQL, MongoDB, MySQL, Redis, Kafka, Snowflake, AWS, Docker, Kubernetes, GitHub Actions, Terraform

Engineering Quality: Code reviews, architecture reviews, API design reviews, test planning, CI/CD, release readiness, Agile/Scrum

PROFESSIONAL EXPERIENCE

JPMorgan Chase & Co.

Jan 2026 – Present

Full-Stack Software Engineer – Graduate Internship, Applied AI & Banking Automation | United States

- Built summary-generation and case-retrieval services within an AI fraud investigation platform, helping analysts review suspicious activity, compare historical cases, and draft investigation notes faster.
- Developed AI-driven chatbot services across web and mobile support flows using RAG, semantic retrieval, and validated prompt workflows, contributing to a ~40% reduction in manual support effort.
- Designed backend services and secure data-access patterns for fraud events, customer profiles, transaction intelligence, and support knowledge using Node.js, TypeScript, Python, PostgreSQL, Redis, and GraphQL.
- Hardened AI workflows with OAuth2/JWT, RBAC, and prompt-validation guardrails that filtered malformed queries before retrieval, and raised delivery quality through architecture, code, and API contract reviews.

FEATURED AI PLATFORM PROJECT

EESA.AI – Live Multi-Tenant AI Agent and Automation Platform

eesa.ai

Next.js, React, TypeScript, MCP, OpenAPI 3.0, Qdrant, Stripe, Docker, Coolify

- Architected a tenant-aware AI agent SaaS with admin, tenant, onboarding, and marketing surfaces organized around protected workflows, plan-aware features, and tenant-specific configuration.
- Designed a 4-layer LLM routing system (master, conversation, orchestration, workflow) with capability-gated model selection, tenant overrides, inheritance, and per-layer cost attribution.
- Built centralized RBAC and permissions infrastructure powering navigation, protected routes, approval visibility, tenant limits, and role-specific feature access from one source of truth.
- Developed an OpenAPI 3.0-to-MCP generator that converts REST API specifications into reusable MCP tool definitions, exposing approved APIs as agent-accessible tools with controlled publishing.
- Shipped and operate the platform end to end — Stripe billing, plugin marketplace with Monaco-based authoring, drag-and-drop dashboards with live MCP tool output, approval queues, and automated Coolify deployments.

ADDITIONAL FULL-STACK EXPERIENCE

TatvaSoft

Jan 2020 – Jul 2023

Full-Stack Software Engineer – Platform Engineering | Nepal / Remote

- Improved platform throughput by 35% by engineering high-volume e-commerce workflows across checkout, orders, inventory, payments, and fulfillment using React.js, Node.js, and PostgreSQL.
- Reduced API response time by 28% by designing secure REST services for payment, order, shipping, and fulfillment integrations with robust validation and error handling.
- Reduced transaction failures by strengthening Stripe and shipping API integrations and hardening order-processing edge cases across checkout.
- Built responsive, accessible UI components and optimized data-access flows; delivered via OAuth2/JWT auth, Dockerized services, and AWS EC2/S3 within Agile sprints and code reviews.

ADDITIONAL AI AND SYSTEMS PROJECTS

Enterprise AI Contract Intelligence Platform

Spring Boot, FastAPI, OpenAI, LangChain, FAISS, Airflow, Spark, Snowflake

- Built an AI-powered contract intelligence system for clause extraction, semantic search, and legal knowledge retrieval using LLMs, RAG, and vector search, with Airflow/Spark/Snowflake pipelines for large-scale ingestion and enterprise reporting.

Real-Time Logistics Visibility Platform

Java, Spring Boot, Kafka, Redis, MongoDB, WebSockets, GitHub Actions

- Engineered an event-driven shipment tracking platform using Kafka, Redis, WebSockets, and microservices for real-time logistics updates, with asynchronous services for inventory sync, monitoring, and CI/CD-backed releases.

PUBLICATIONS AND RESEARCH

- MST: A Modified Sparse Transformer with depth-aware attention for multi-modal camera-LiDAR fusion in autonomous vehicles — Transportation Research Interdisciplinary Perspectives, Vol. 34, Article 101571, 2025.
- Advancing Clinical Translation in Nepali through Fine-Tuned Multilingual Models — Proceedings of RANLP 2025, pp. 48-56, Varna, Bulgaria, 2025.

EDUCATION

California State University, Dominguez Hills <i>Master of Science in Computer Science Carson, CA</i>	Aug 2024 - May 2026
London Metropolitan University / Informatics College Pokhara <i>Bachelor of Science in Computing Pokhara, Nepal</i>	Aug 2018 - Dec 2021